

AIM:

To implement the simple substitution technique named Caesar cipher using C language.

ALGORITHM:

STEP-1: Read the plain text from the user.

STEP-2: Read the key value from the user.

STEP-3: If the key is positive then encrypt the text by adding the key with each character in the plain text.

STEP-4: Else subtract the key from the plain text.

STEP-5: Display the cipher text obtained above.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main() {    char plain[100],  
cipher[100];    int key, i, length;
```

```
    printf("Enter the plain text: ");    scanf("%s", plain);
```

```
    printf("Enter the key value: ");    scanf("%d", &key);
```

```
    key = key % 26;
```

```
    length = strlen(plain);
```

```
printf("\nPLAIN TEXT    : %s", plain);    printf("\nENCRYPTED TEXT
: ");
```

```
for(i = 0; i < length; i++)
{
    if(isupper(plain[i]))
    {
        cipher[i] = ((plain[i] - 'A' + key) % 26) + 'A';
    }
    else if(islower(plain[i]))
    {
        cipher[i] = ((plain[i] - 'a' + key) % 26) + 'a';
    }
    else {
cipher[i] = plain[i];
    }
}
```

```
printf("%c", cipher[i]);
}
```

```
cipher[length] = '\0';
```

```
printf("\nDECRYPTED TEXT : ");
```

```
for(i = 0; i < length; i++)
{
    if(isupper(cipher[i]))
    {
        plain[i] = ((cipher[i] - 'A' - key + 26) % 26) + 'A';
    }
    else if(islower(cipher[i]))
    {
        plain[i] = ((cipher[i] - 'a' - key + 26) % 26) + 'a';
    }
    else {
plain[i] = cipher[i];
    }
}
```

```
}

printf("%c", plain[i]);

}

plain[length] = '\0';

return 0;}
```

OUTPUT:

Output Clear

Enter the plain text: run
Enter the key value: 2

PLAIN TEXT : run
ENCRYPTED TEXT : twp
DECRYPTED TEXT : run

=== Code Execution Successful ===

RESULT:

Thus the implementation of Caesar cipher had been executed successfully.